# PAOLO VIVIANI


## PhD – Research Engineer

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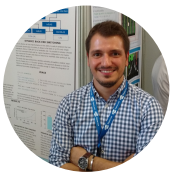

## EDUCATION

### M.Sc. in Theoretical physics

**University of Torino, 104/110**

2015  Collegio Universitario "R. Einaudi", Torino

Scholarship winner: "Piano Lauree Scientifiche 2008", granted by Società Italiana di Fisica.

### Ph.D. in Computer Science

**University of Torino, Computer Science Dept.**

2015 – 2019  Torino, IT

Thesis: *Deep Learning at Scale with Nearest Neighbours Communications*. Supervisor: Marco Aldinucci. Funded by Noesis Solutions.

- Co-supervisor of one M.Sc. Thesis

## EXPERIENCE

### Noesis Solutions

**Reasearch Engineer**

April 2015 – Permanent  Novara, IT

- Development of **machine learning modelling methodologies for engineering applications**.
- Re-definition of the internal stack (C++, Python) for scientific computing, with focus on performance and maintainability.
- Cloud and container technologies exploitation.
- Supervisor of one internship.
- Re-designed company-wide source code management workflow.
- **Technical lead for the following research projects**. Non-technical tasks included presentations to European Commission reviewers and reporting activities.

### Funded Research Projects

**MACH – ITEA2 project**

April 2015 – December 2016  itea3.org/project/mach.html

**Fortissimo 2 – H2020-FoF project**

February 2016 – October 2018  www.fortissimo-project.eu

**CloudFlow – FP7-I4MS project**

March 2016 – May 2017  www.eu-cloudflow.eu

**Blockchain for online Service Security – IMEC-ICON Flemish project**

March 2018 – November 2019

**VaProFam – FlandersMake-ICON Flemish project**

February 2019 – ongoing

**DC-CDS – FlandersMake-ICON Flemish project**

April 2019 – ongoing

## FIELDS OF INTEREST

- High Performance Computing
- Machine Learning
- Containers and Cloud technologies

## SKILLS

| | |
|---|---|
| Parallel Computing | ●●●●● |
| GPU Computing | ●●●○○ |
| Cloud technologies | ●●●○○ |
| Machine Learning | ●●●●○ |

### Programming

| | |
|---|---|
| C++ | ●●●●● |
| Python | ●●●●○ |
| CUDA | ●●○○○ |
| Fortran | ●●●○○ |
| Git | ●●●●● |

### Soft

| | |
|---|---|
| Presentations | ●●●●○ |
| Communication of results | ●●●●● |
| Factorization of problems | ●●●●● |
| Formalization of requirements | ●●●●○ |
| Time management | ●●○○○ |
| Workarounds | ●●●●● |

### Tools

- Keras
- MxNet
- Pandas
- Apache Spark
- Openstack
- Docker
- Gitlab CI
- Linux
- IP Networking
- Latex
- Mathematica

CISCO CCNA course completed in 2008.

## LANGUAGES

| | |
|---|---|
| Italian | ●●●●● |
| English | ●●●●● |
| French | ●●○○○ |

# ACADEMIC

## Publications

1. Reniers, V., Gao, Y., *et al. Authenticated and Auditable Data Sharing via Smart Contract* in *Proceedings of the 35th ACM/SIGAPP Symposium on Applied Computing* (ACM, Brno, Czech Republic, 2020), 1–8.
2. Drocco, M., Viviani, P., Colonnelli, I., Aldinucci, M. & Grangetto, M. *Accelerating spectral graph analysis through wavefronts of linear algebra operations* in *Proc. of 27th Euromicro Intl. Conference on Parallel Distributed and network-based Processing (PDP)* (IEEE, Pavia, Italy, 2019), 9–16.
3. Reniers, V., Van Landuyt, D., *et al. Analysis of Architectural Variants for Auditable Blockchain-based Private Data Sharing* in *Proceedings of the 34th ACM/SIGAPP Symposium on Applied Computing* (ACM, Limassol, Cyprus, 2019), 346–354.
4. Viviani, P. *Deep Learning at Scale with Nearest Neighbours Communications* PhD thesis (Computer Science Department, University of Torino, Sept. 2019).
5. Viviani, P., Drocco, M., Baccega, D., Colonnelli, I. & Aldinucci, M. *Deep Learning at Scale* in *Proc. of 27th Euromicro Intl. Conference on Parallel Distributed and network-based Processing (PDP)* (IEEE, Pavia, Italy, 2019), 124–131.
6. Aldinucci, M. *et al. HPC4AI, an AI-on-demand federated platform endeavour* in *ACM Computing Frontiers* (Ischia, Italy, May 2018).
7. Tordini, F., Aldinucci, M., Viviani, P., Merelli, I. & Liò, P. *Scientific Workflows on Clouds with Heterogeneous and Preemptible Instances* in *Proc. of the Intl. Conference on Parallel Computing, ParCo 2017, 12-15 September 2017, Bologna, Italy* (IOS Press, 2018).
8. Viviani, P., Drocco, M. & Aldinucci, M. Pushing the boundaries of parallel Deep Learning - A practical approach. *CoRR* **abs/1806.09528** (2018).
9. Viviani, P., Drocco, M. & Aldinucci, M. *Scaling Dense Linear Algebra on Multicore and Beyond: a Survey* in *Proc. of 26th Euromicro Intl. Conference on Parallel Distributed and network-based Processing (PDP)* (IEEE, Cambridge, United Kingdom, 2018).
10. Viviani, P., Aldinucci, M., d'Ippolito, R., Lemeire, J. & Vucinic, D. in *Improved Performance of Materials: Design and Experimental Approaches* 93–106 (Springer International Publishing, Cham, 2018).
11. Viviani, P., Torquati, M., Aldinucci, M. & d'Ippolito, R. *Multiple back-end support for the Armadillo linear algebra interface* in *In proc. of the 32nd ACM Symposium on Applied Computing (SAC)* (Marrakesh, Morocco, Apr. 2017), 1566–1573.
12. Viviani, P. *Parallel Computing Techniques for High Energy Physics* MA thesis (Physics Department, University of Torino, 2015).

## Other

- Several PhD schools attended
- Program Committee member, *Euromicro International Conference on Paralle, Distributed, and Network-based Processing (PDP)* for 2018 (also session chair), 2019 and 2020.
- Program Committee member, *Parallel Numerical Methods and Libraries for Heterogeneous Multi/Manycores (PDP2018 and PDP2019)*
- Program Committee member, *Artifact Evaluation, Euro-Par 2018*
- Program Committee member, *16th IEEE International Conference on Scalable Computing and Communications (ScalCom 2016)*

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# FREE TIME

Avid traveller and photographer.
Former track & field athlete, now turned to playground basketball.

paoloph.myportfolio.com

### Photographer

**OASIS Project**

2013 — Kharga, Egypt

Archaeological expedition funded by Fondazione Collegio delle Università Milanesi and American University in Cairo.

### Photographer

**Collegio Universitario "R. Einaudi"**

2012 – 2014 — Torino, Italy

### Exchange student

**Tauranga Boy's College**

2006 — Tauranga, New Zealand

### Scuba Diver

**PADI Advanced Open Water Diver**

From 2004

### Track and Field athlete

**Javelin Throw – Regional and National competitions**

1997 – 2012

### Volunteer

**IAAF Athletics World Championship**

2009 — Berlin, Germany

### English Course

**'General English Super Intensive' C1 level course at Alpha College of English**

2012 — Dublin, Ireland